

(TR-104) PYTHON WITH AI/ML

Training Day 105 Report

Date: 5 June 2026

Topic Covered: Revision of Training Concepts, Project Workflows & AI/ML Technologies

The hundred and fifth day of training focused on revising the concepts, technologies, workflows, and project modules covered throughout the training program to strengthen understanding and prepare for project evaluation and viva discussions.

- Revised major concepts covered during the Python with AI/ML training program.
- Reviewed AI and Machine Learning fundamentals along with their practical applications.
- Revisited concepts related to Generative AI, Prompt Engineering, Large Language Models (LLMs), and AI Agents.
- Revised LangChain concepts including chains, tools, agents, prompts, and workflow orchestration.
- Reviewed Retrieval-Augmented Generation (RAG) architecture, document retrieval, vector databases, and question-answering workflows.
- Revisited FastAPI concepts including API development, routing, request handling, and backend integration.
- Reviewed MongoDB concepts including collections, documents, aggregation pipelines, indexing, and query optimization.
- Revised AI Attendance System workflows including face recognition, voice recognition, attendance processing, and authentication mechanisms.
- Reviewed image recognition concepts including face detection, facial embeddings, similarity matching, and classification workflows.
- Revisited voice recognition concepts including voice embeddings, speaker identification, and verification processes.
- Reviewed Streamlit application architecture, session management, frontend-backend integration, and deployment workflows.
- Studied AI Assistant architecture including tool calling, agent workflows, database interaction, and conversational query handling.

- Revised project documentation, system design concepts, software development lifecycle, testing methodologies, and report preparation techniques.

Overall, the training strengthened understanding of the concepts, technologies, tools, and project workflows covered throughout the internship, improving readiness for project demonstrations, documentation reviews, and viva evaluations.